

Light sink float separation - Proof of concept

Testing Document

Version 1

Date:

- Start: 03.07.2025
- End: 07.07.2025

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General description and purpose

Purpose of test

By combining the results of test 1 “[Sink-float separation – Proof of concept](#)” which grouped the mixed plastic waste in low-density polymers (LDP) – PP, LDPE, HDPE – and high-density polymers (HDP) – PS, PET, PVC – with the results of test 3 “[LDP separation – Proof of concept](#)” which extracted the LDPE foils from the LDP mix, a blend of PP and HDPE remains. This test assesses the separation of these two types by sink-float separation using an ethanol-water mixture as a density-adjusted separation medium. Additionally, the evaporation of the medium over time is investigated as well as the thread of ignition when using ethanol.

Description of Test

For this test, the color-coded plastic flakes from test 1 are used. One mixed test plastic batch is created from the PP and the HDPE flakes, using 1.5 g of each material.

The densities of the materials are 0.900-0.910 g/cm³ for PP and 0.940-0.970 g/cm³ for HDPE (SpecialChem SA, 2025) which lie within the density boundaries of the separation solution (100% ethanol: ρ^{20} 0.78932 g/cm³, 100% water: ρ^{20} 0.99820 g/cm³ (Rumble, 2025b)). As a separation medium, an ethanol-water solution is prepared for which a temperature of 20-25 °C is assumed. At this temperature range, the interesting ethanol mass % range is shown in Table 1.

Table 1: Density of Ethanol–Water Mixtures at different percentages and temperatures (interpolated values in red) (Rumble, 2025b)

Ethanol mass %	ρ^{20} [g/cm ³]	ρ^{25} [g/cm ³]	Ethanol volume %
35	0.94491	0.94143	40.5
40	0.93515	0.93145	45.7
44.13	0.92650	0.92266	50
45	0.92469	0.92082	50.9
50	0.91381	0.90982	55.8
90	0.81795	0.81360	91.9
94	0.80721	0.80263	95.2
95	0.80422	0.79989	96.0

To leave margin for inaccuracy and account for some evaporation of the ethanol while simultaneously minimizing the required amount of ethanol, an ethanol volume % of 50 % is targeted.

The ethanol used is “Ethanol denatured with Ketone 94 %” from the HCI-Shop, ETH Zurich. Since the composition is not further specified, it is assumed that it consist of 94 % ethanol and 6 % methyl ethyl ketone (MEK) which is the ketone commonly used for denaturing in the EU (“Vergällung,” 2024). Since the density of MEK (ρ^{25} 0.7999 g/cm³ (Rumble, 2025a)) is similar to the one of ethanol, the simplification is made that the mixture consists of 100% ethanol.

To create the separation medium of ethanol-water mixture with an ethanol volume % of 50 %, equal volumes of ethanol and water (150 ml each) are added to a beaker. To verify the ethanol content with

the hydrometer, 100 ml of the mixture are transferred to a measuring cylinder. After recording the value, the liquid is returned to the beaker. Then, the mixed plastic test batch is added and stirred with a magnetic mixer. After stirring for 3 minutes at 300 RPM, the magnetic mixer is turned off to allow the material to settle for 2 minutes. The floating fraction is skimmed with a strainer. The same is done for the sink fraction while transferring the separation solution to another beaker and both extracted fractions are placed on paper towels to dry. Subsequently, the fractions are sorted into the plastic types to measure the masses and calculate the purities.

To assess the evaporation rate of the ethanol under different conditions, the separation solution is then transferred to two identical measuring cylinders (100 ml each). After the alcohol content and mass is recorded with the hydrometer for each batch, one of the cylinders is sealed with a chemically resistant glove. The alcohol content and mass are recorded for both batches three more times.

Lastly, it is attempted to light a small quantity of the separation medium (50 vol % ethanol) as well as a small quantity of the denatured ethanol (94 vol % ethanol) with a match to test the hazard of ethanol ignition at different ratios.

Calculations

Conversion from ethanol mass % (a) to ethanol volume % (b):

$$\begin{aligned} \% w/w: a &= \frac{m_E}{m_W + m_E} \rightarrow \frac{m_E}{m_W} = \frac{a}{1-a} \rightarrow \frac{V_E}{V_W} = \frac{a}{1-a} \frac{\rho_W}{\rho_E} \\ \% v/v: b &= \frac{V_E}{V_W + V_E} \rightarrow \frac{V_E}{V_W} = \frac{b}{1-b} \\ \frac{a}{1-a} \frac{\rho_W}{\rho_E} &= \frac{b}{1-b} \Rightarrow b = \frac{1}{\frac{1-a}{a} \frac{\rho_E}{\rho_W} + 1} \end{aligned}$$

Plastic separation results:

$$\text{Purity } \varphi_{float}: \quad \varphi_{float} = m_{float_{PS}} / m_{float}$$

$$\text{Purity } \varphi_{sink}: \quad \varphi_{sink} = m_{sink_{PET+PVC}} / m_{sink}$$

$$\text{Purity } \varphi_{combined}: \quad \varphi_{combined} = (\varphi_{sink} + \varphi_{float}) / 2$$

Schedule of test with all steps required

1. Prepare batch of mixed test plastics (**PP: 1.5 g, HDPE: 1.5 g**).
2. Prepare the separation solution in a beaker with **150 ml** of ethanol and **150 ml** of tap water.
3. Transfer **100 ml** of the solution to a measuring cylinder.
4. Record the ethanol content with the hydrometer.
5. Return the solution to the beaker.
6. Add a magnetic stir bar and place the container on a magnetic mixer.
7. Set up camera to record the separation behavior over the test timeframe and start recording.
8. Start the mixer at **300 rpm** for 3 minutes. Add mixed test plastic batch to the test container.
9. After 3 minutes, turn of the magnetic mixer and let the plastic flakes settle for 2 minutes.

10. Use the strainer to collect the float fraction and place on a paper towel to dry.
11. Pour the content of the container through the strainer into another beaker to collect the separation solution and to remove the sink fraction.
12. Place the sink fraction on a paper towel to dry.
13. After drying off the flakes, sort the float and sink fractions by color and measure the masses.
14. Transfer 2x **100 ml** of the separation solution in two identical measuring cylinders.
15. Record the masses and the ethanol contents of each batch.
16. Seal one of the cylinders with a chemically resistant glove.
17. Repeat steps **15** and **16** after 1.5, 5, and 11 hours.
18. Prepare a small quantity of the separation solution in a beaker.
19. Set up the camera to record ignition and combustion and start recording.
20. Light a match and drop the match into the filled beaker.
21. Once the flame has extinguished, wait until the glass of the beaker has returned to room temperature and remove any leftover liquid inside.
22. Repeat steps **18-21**, substituting the separation solution with the denatured ethanol.

Risk analysis

Potential risks associated with this test

Ethanol is highly flammable in liquid and vapor form and can cause serious eye irritation.

Mitigation steps

Only small amounts of ethanol is used at a time, the tests are done in a ventilated area, possible sources of fire are removed, and a fire blanket is prepared.

Safety glasses are worn at any time.

For the ignition tests, only small quantities are used, and the test is performed in a large space with any other flammable material removed.

Risks after taking mitigation steps

None.

Tools and devices required

Tool	Check-in tick
1x safety glasses	X
1x fire blanket	X
Paper towels	X
1x laptop	X
1x camera (phone)	X

1x timer (laptop or phone can be used)	X
1x scale	X
Containers of cut plastic flakes (PP, HDPE)	X
1x container for mixed plastic test batches	X
2x transparent beaker	X
2x measuring cylinders	X
1x hydrometer	X
1x magnetic stir bar	X
1x magnetic mixer	X
1x strainer	X
1x matchbox	X
150 ml tap water	X
150 ml ethanol	X
1x nitrile rubber glove	X
1x tweezers	X
1x calculator	X

Testing Results

All recordings and pictures of the tests are available [here](#).

Plastic separation

Mass of beaker: 140.5 g.

Test	Input masses [g]		Float-output masses [g]		Sink-output masses [g]	
a (160 rpm)	PP 1.5	HDPE 1.5	PP 1.5	HDPE 0	PP 0	HDPE 1.5
	Tot: 3.0 (theoretically 3.0)					

Any unexpected behaviors:

- No residual of separation solution measurable on plastic flakes after drying

Ethanol evaporation

Mass of measuring cylinders: 41.7 g each.

Batch	Time of measurement	Total mass [g] (with cylinder)	Ethanol content [vol %]
Initial solution	10:53	133.7 ¹	52
a (sealed)	11:11	134.4	48
b	11:11	134.4	48
a (sealed)	12:46	134.1	47
b	12:46	133.9	47
a (sealed)	16:06	133.8	46
b	16:06	133.6	46
a (sealed)	22:49	133.6	46
b	22:49	133.3	46
a (sealed)	15:00 (+3 days)	133.1	46
b	15:01 (+3 days)	131.3	45

Ethanol ignition

At 50 vol % ethanol, the solution was not ignited by the match. Instead, the match extinguished immediately when coming in contact with the solution. With an alcohol content of 94 vol %, the denatured ethanol exhibited a long lasting, almost invisible flame.

¹ The liquid was transferred back into the beaker to conduct the plastic separation test after this measurement. For the next observations in this column, the measuring cylinders were filled anew, resulting in a different starting weight.

Evaluation

The results obtained for the sink-float separation show an ideal separation of the PP flakes (float fraction) from the HDPE flakes (sink fraction) (see Table 2).

Table 2: Purity results sorted by combined purity.

Purity float	Purity sink	Purity combined
100.00%	100.00%	100.00%

The evaporation test shows a decrease of mass and ethanol content of the separation solution over time. While only small in this case, the evaporation rate is dependent on the surface area (Carrier et al., 2016). Therefore, sealing is recommended for larger containers.

Although the solution at 50 vol % ethanol could not be ignited during the testing, caution should still be exercised when handling ethanol. All sources of ignition should be avoided in the area of use.

Bibliography

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